

Consumer Sentiment & Machine Learning Analytics for Bambara Groundnut (BGN) Beverage Innovation

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Executive Portfolio KPI Dashboard

97.1% Bamilk Acceptance 52.2% Very Good 23.2% Excellent	97.6% BamPro Acceptance 54.2% Very Good 21.7% Excellent	87.1% Bamilk Purchase Intent 45.7% Probably 41.4% Definitely	82.3% ML Model Accuracy Naïve Bayes & Random Tree	97.9% Positive Recall Rate 10-Fold Cross-Validation
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PORTFOLIO HIGHLIGHT: COMMERCIAL & SCIENTIFIC FEASIBILITY

Consumer sentiment analysis of Bambara groundnut (BGN) plant milk (Bamilk) and cultured beverage (BamPro) confirms an overwhelmingly positive response (>97% rating average-to-excellent). Advanced Natural Language Processing (NLP) and Machine Learning models (WEKA implementation) achieved 82.29% classification accuracy in predicting consumer sentiment, providing a robust virtual screening framework for ethnic food commercialization.

1. Research Objectives & Strategic Context

Bambara groundnut (BGN, *Vigna subterranea*) is an underutilized, drought-tolerant African legume rich in plant protein, low in fat, and abundant in antioxidant phenolics. Despite its high therapeutic and functional food value, commercial application has traditionally been restricted to simple household boiling or roasting. This research portfolio presents the commercial pre-feasibility, consumer acceptability, and machine learning sentiment evaluation of two novel value-added BGN products showcased at the Cape Peninsula University of Technology (CPUT) Innovation Showcase:

- Bamilk:** A smooth, nutrient-dense plant-based milk alternative derived from processed BGN seeds, formulated as an affordable dairy substitute for lactose-intolerant consumers.
- BamPro:** A cultured probiotic fermented beverage (yoghurt analogue) developed from BGN milk, engineered to offer digestive health benefits and enhanced shelf stability.

The primary objectives of this Technology Innovation Agency (TIA) Seed Fund project were:

1. Evaluate consumer sensory perception, quality/taste ratings, and willingness to purchase Bamilk and BamPro across diverse demographic groups.

2. Establish Generalized Linear Estimating Models (GLM) to quantify the specific impacts of age, gender, and race on purchase intention.
3. Develop Natural Language Processing (NLP) text-mining pipelines and WEKA machine learning classifiers to automatically categorize consumer sentiments and extract key sensory drivers for product optimization.

2. Methodology & Machine Learning Data Pipeline

The research executed an integrated empirical sensory evaluation combined with automated machine learning sentiment classification:

Pipeline Stage	Methodology & Analytical Implementation
Data Collection & Sampling	103 consumers (54 male [52.4%], 49 female [47.6%]) evaluated Bamilk (71 tasters) and BamPro (84 tasters) at CPUT Innovation Showcase. A 1-page structured questionnaire captured demographics, quality/taste ratings, purchase likelihood, and open-ended improvement comments.
Statistical & GLM Analysis	Data expressed as frequencies and percentages. Chi-square (χ^2) tests determined proportion significance. Generalized Linear Estimating Models (GLM in IBM SPSS v23) evaluated main effects of gender, age, and race on purchase intention.
NLP Text Pre-processing	Consumer comments mined via WEKA 3.8.0. Unigrams extracted using StringToWordVector filter. Class imbalance resolved via Synthetic Minority Oversampling TEchnique (SMOTE) to equalize positive and negative instances (48 each, 96 total).
Feature Engineering & Reduction	Stop-words removed; Porter's stemming applied via Snowball stemmer. AttributeSelect filter reduced initial 34 descriptors to 9 core predictive unigrams: bit, flavour, improve, milk, thicker, acidic, bampro, market, and product.

3. Results & Empirical Discussion

Table 1: Demographic Distribution of Survey Respondents (N = 103)

Gender	< 21	21–25	26–30	31–35	36–40	> 40	Total
Male	1 (1.9%)	13 (24.1%)	6 (11.1%)	10 (18.5%)	1 (1.9%)	23 (42.6%)	54 (100%)
Female	2 (4.1%)	15 (30.6%)	5 (10.2%)	5 (10.2%)	5 (10.2%)	17 (34.7%)	49 (100%)
Total	3 (2.9%)	28 (27.2%)	11 (10.7%)	15 (14.6%)	6 (5.8%)	40 (38.8%)	103 (100%)

Sensory Rating & Purchase Likelihood: Consumer response to both BGN beverages was overwhelmingly positive. **Bamilk** achieved a 97.1% combined rating from average to excellent (52.2% very good, 23.2% excellent, 21.7% average; $\chi^2 = 59.91$, $p < 0.05$). **BamPro** achieved a 97.6% combined rating (54.2% very good, 21.7% excellent, 21.7% average; $\chi^2 = 78.14$, $p < 0.05$). Purchase intention ranged from probably to definitely buy for 87.1% of Bamilk respondents ($\chi^2 = 65.57$, $p < 0.05$) and 85.2% of BamPro respondents ($\chi^2 = 70.42$, $p < 0.05$). The 25th percentile for both products scored 'probably buy', indicating that over 75% of consumers expressed definitive purchase intent.

Demographic Predictors (GLM Models): Generalized Linear Estimating Models in SPSS established that **age and race significantly ($p < 0.05$) influenced purchase intention**, whereas gender had no significant effect ($p > 0.05$). Consumers aged > 40 exhibited higher likelihood to purchase Bamilk, whereas consumers < 21 showed higher affinity for BamPro. Across racial profiles (46.5% Black, 29.7% Coloured, 20.8% White, 3.0% Asian), Black and Indian/Asian respondents displayed the highest purchase intent for Bamilk, while Black and Coloured respondents preferred BamPro. This reflects cultural familiarity with Bambara groundnut and a consumer desire for modern, convenient beverage formats beyond traditional boiled seeds.

Qualitative Attribute Profiling: Dominant comment descriptors included *flavour, milk, bit, thicker, acidic, improve, and sugar*. Qualitative analysis revealed key consumer sensory recommendations: (1) adding 'a bit' of sugar to balance acidity in BamPro yoghurt, (2) increasing BamPro consistency to a thicker viscosity, and (3) replacing artificial strawberry flavor with natural flavorings to avoid masking BGN's natural pleasant nutty taste.

4. Machine Learning Model Evaluation & Virtual Screening

Four supervised machine learning classification algorithms (J48, Random Forest, Random Tree, and Naïve Bayes) were trained and validated on the 96 pre-processed comment instances using 10-fold stratified cross-validation:

Table 2: Machine Learning Model Evaluation for BGN Sentiment Classification

Algorithm	Class	TP Rate	FP Rate	Precision	Recall	F-Measure	ROC Area
J48 (C4.5)	Negative	0.646	0.021	0.969	0.646	0.775	0.754
J48 (C4.5)	Positive	0.979	0.354	0.734	0.979	0.839	0.754
Random Forest	Negative	0.646	0.021	0.969	0.646	0.775	0.840
Random Forest	Positive	0.979	0.354	0.734	0.979	0.839	0.840
Random Tree	Negative	0.646	0.021	0.969	0.646	0.775	0.829
Random Tree	Positive	0.979	0.354	0.734	0.979	0.839	0.829
Naïve Bayes	Negative	0.667	0.021	0.970	0.667	0.790	0.838
Naïve Bayes	Positive	0.979	0.333	0.746	0.979	0.847	0.830

Model Analysis & Decision Rules: Overall classification accuracy reached **82.29% for Naïve Bayes and Random Tree** (81.25% for J48 and Random Forest), with Kappa statistics ranging from 0.625 to 0.646. All four algorithms demonstrated outstanding positive sentiment recall (97.9%), correctly classifying 47 out of 48 positive consumer comments. The high ROC area values (> 0.82 for RF, RT, and NB) confirm strong discriminative capability. In decision tree analysis, *flavour* emerged as the root attribute with highest information gain.

Virtual Screening Cost-Benefit Analysis: Cost-benefit simulation indicated that utilizing Naïve Bayes or Random Tree for virtual sentiment screening requires selecting only **65.6% of the consumer pool** to achieve 97.9% positive sentiment capture, delivering a cost score of 17 and a net gain score of 31. This validates machine learning as an efficient tool for screening consumer feedback in food product development pipelines.

5. Strategic Recommendations & Action Plan

PORTFOLIO ACTION PLAN: COMMERCIALIZATION ROADMAP

1. Product Reformulation: Adjust BamPro culture fermentation parameters to moderate acidity, increase viscosity, and incorporate natural sucrose sweetening.
2. Flavor Standardization: Remove artificial strawberry flavoring from Bamilk; emphasize BGN's clean, rich, nutty profile.
3. Market Targeting: Focus retail placement on urban lactose-intolerant demographics, top LSM retail channels, and young consumers (<21 for BamPro, >40 for Bamilk).
4. Scale-up & Virtual Screening: Deploy Naïve Bayes NLP pipeline to monitor continuous consumer feedback during commercial pilot distribution.